4. (*i*) $5m$ (*ii*) $3p^3q^2$ 5. (*i*) $16y-4z$ (*ii*) $2mn^2+8m^3n-\frac{1}{2}$ (*iii*) $\frac{5}{6}y-3xy^2+1$
(*iv*) $=9p^2r+18pq-45$

6. (*i*) $9y^2$ (*ii*) $9xy$ (*iii*) $4m^2-3m$ (*iv*) $16n^2-2n+3$ (*v*) x^2+1x-6 (vi) $-(3p^2-4p+7)$

7. (ii) A is true but B is false 8. (i) both A and B are true

Exercise 3.3

1. (*i*) $9m^2+30m+25$ (*ii*) $25p^2-10p+1$ (*iii*) $4n^2+4n-3$ (*iv*) $(2p+5q)(2p-5q)$

2. (*i*) $m^3+9m^2+27m+27$ (*ii*) $8a^3+60a^2+150a+125$ (*iii*) $27p^3+108p^2q+144pq^2+64q^3$
(iv) 14,0608 (v) 1124864

3. (*i*) $125-75x+15x^2-x^3$ (*ii*) $8x^3-48x^2y+96xy^2-64y^3$ (*iii*) $a^3b^3-3a^2b^2c+3abc^3-c^3$
(iv) 110, 592 (*v*) $912673x^3y^3$

4. p^3-5p^2+2p+8 5. x^3+3x^2+3x+1 6. x^3-2x^2-5x+6

7. (C) 2 8. (B) a^2+ab+b^2 9. (A) p^3+q^3 10. (D) 72 11. (D) $3ab(a+b)$

Exercise 3.4

1. (*i*) $6y(3x-2z)$ (*ii*) $3x^2y(3x^3y^2+2xy-6)$ (*iii*) $(b-2c)(x+y)$
(*iv*) $(x+y)(a+b)$ (*v*) $(4x-1)(2x^2-1)$ (*vi*) $(x-2)[3y(x-2)+2]$
(*vii*) $2y(9x-2y-zx)$ (*viii*) $(a-3)(a^2+1)$ (ix) $3y(y+4)(y-4)$ (x) $(b-1)(ab-c^2)$

2. (*i*) $(x+7)^2$ (*ii*) $(y-5)^2$ (*iii*) $(c+2)(c-6)$
(*iv*) $(m+9)(m-8)$ (*v*) $(2x-3)(2x-1)$ 3. (i) $(4x+3)^3$ (ii) $(3p+2q)^3$

4. (i) $(y-6)^3$ (ii) $(2m-5n)^3$ 5. (D) $3x$, $(3x+2y)$ 6. (C) $(2+m)(2-m)$

7. (D) x^2-x-20 8. (B) 3 9. (C) $(1-m)(1+m+m^2)$ 10. (B) $(x+y)$

Exercise 3.5

Miscellaneous Practice Problems

1. $7x^2y^5+4x^4y^3+60x^2y^2$ 2. $12x^3-8x^2+27x-18$ 3. $S.I=\frac{7}{5}a^3b^4$
4. $\frac{1}{2}a+2b+4$ 5. $(y-3)(7y+2)$

Challenging Problems

6. $4x+3$ 7. $x+2$ 8. $(y+7)(y+8)$ 9. $(4p^2+1)(2p+1)(2p-1)$ 10. $3(x-5y)^3$

Exercise 3.6

1. (i) $x=7$ (ii) $y=11$ (iii) $m=7$ (iv) $p=15$ (v) One
2. (i) True (ii) False 3. (c) (iii),(i), (iv), (v), (ii)
4. (i) $x=11$ (ii) $y=\frac{1}{-4}$ (iii) $x=-1$ 5. (i) $x=-4$ (ii) $p=-1$ (iii) $x=-11$
6. (i) $x=-2$ (ii) $m=-4$

Exercise 3.7

1. (i). $x=-\frac{b}{a}$ (ii) Positive (iii) $x=30$ (iv) 40° (v) b=9
2. (i) True (ii) False (iii) False 3. 3, 21 4. 27
5. $l=8$ cm, $b=24$ cm 6. (80,10) notes 7. Murali's is 15 years old, Thenmozhi's age is 20 years old 8. 63 9. $\frac{13}{21}$
10. 37.4km 11. (B) 20 12. (A) 62° 13. (C) 10000 14. (C) 4 15. (D) $(x-1)$

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